Ahmedabad University

CSE 631: Machine Learning

Weekly Report(5)

(Group - 9) Learning Machine

Section 1

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Weekly Progress Report - 5

Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

1. Detailed Summary of Work Done During the Current Week

The fifth project week was dedicated to the discussion of the classical method of machine learning to detect and to count decorative stones on embroidery pictures. Rather than resorting to using deep learning models directly, such as YOLO, the aim of this step was to apply a more classical computer vision pipeline to gain a better insight into the underlying machine learning principles.

This week was primarily aimed at designing and implementing a Sliding Window + HOG + SVM-based detection system, a system that offers a clear and understandable pipeline on the object detection system. The method focuses on the extraction of features, supervised learning, and organized detection, which is why it is the most appropriate approach to be used in a course of machine learning.

1.1 Motivation for Classical Machine Learning Approach

Even though the deep learning models are very accurate, the classical machine learning techniques offer:

- Better interpretability
- Good understanding of foundational ML.
- Effective knowledge of feature-extraction and classification.
- The pipeline of detection is a step-by-step method to detect tasks and worksheets, each of which will be examined individually subsequently.

The approach that has been chosen comprises:

- OBHOG (Oriented Binary Histogram of gradients) of feature extraction.
- SVM (Support Vector Machine) of classification.
- Sliding Window object detection method.

This combination finds extensive use in conventional computer vision as well as offers a good theoretical foundation.

1.2 Image Preprocessing and Enhancement

Preprocessing methods were employed prior to the extraction of features to enhance the quality of the image and bring out the areas of the stone.

The preprocessing operations involved:

- Image to grayscale conversion.
- CLAHE (Contrast Limited Adaptive Histogram Equalization) Application.
    1. Sharps contrast in local areas.
    2. Enhances the quality of stones in contrast to fabric textures.

This action assists in sorting Jarkans (stones) and sophisticated patterns of embroideries.

1.3 Sliding Window Strategy

An image process that scanned the image to find stone areas using sliding window technique was used.

The process includes:

- Every tile (640x 640) is subdivided into smaller patches (32x 32).
- An image is sliding through a window in a systematic manner.

- Analysis of the patches is done separately.

This method allows:

- Regional calcification of stones.
- Detection of small objects
- General coverage of the whole image.

1.4 Feature Extraction using HOG

The HOG (Histogram of Oriented Gradients) features were computed on each image patch.

HOG captures:

- Edge directions
- Gradient structures
- Figural information of objects.

These characters are efficient in detecting small items such as decoration stones particularly when the background is smooth.

1.5 SVM Classifier Training

Labelled data was used to train a Support Vector Machine (SVM) classifier.

Training process:

- Positive samples:

  The annotated stone regions were extracted using CVAT labels.

- Negative samples:

  Hazardous patches randomly sampled off non-stone areas.

The SVM model is trained to classify:

- Stone (Jarkan)

- Non-stone (background)

This is a learning generation that can supervise the model to differentiate between stones using the extracted features.

1.6 Detection and Non-Maximum Suppression

Upon classification of every patch:

- The number of overlapping detections can be of multiple ones.
- Non-Maximum Suppression (NMS) is used.

NMS helps to:

- Eliminate redundant notifications.
- Only the most confident bounding boxes should be kept.

This enhances precision and readability of the final results of detection.

1.7 Stone Counting and Image Reconstruction

The last step to the pipeline will be:

- Summation of stones found on each patch.
- Backing up the detections to their source locations.
- Raising up the entire fabric image.

The sum of the number of stones is determined as:

- Adding detections of all sliding windows.

This produces:

- Final detection output
- Total number of stones used in the fabric image.

1.8 Visualization of Detection Results

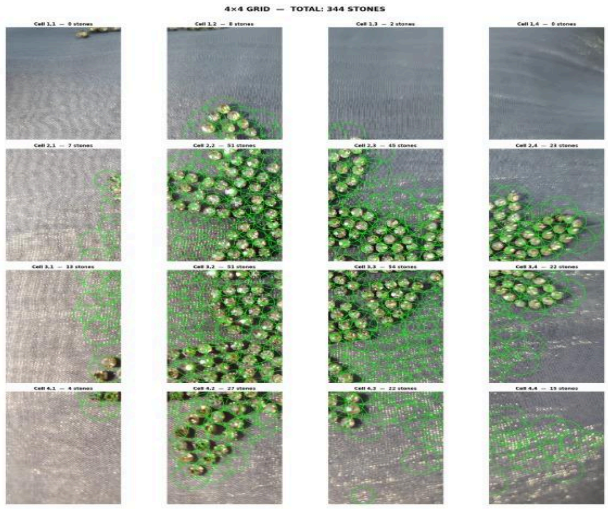
In order to measure the system performance:

- Detection of stone regions is done and bounding boxes are drawn on the regions.
- Sample results are graphically represented.
- Ground truth annotations are compared to results of detection.

This helps in understanding:

- Accuracy of detection
- False detections and false positives.

Image:


4×4 GRID — TOTAL: 344 STONES
Cell 1,1 — 0 stones
Cell 1,2 — 8 stones
Cell 1,3 — 2 stones
Cell 1,4 — 0 stones
Cell 2,1 — 7 stones
Cell 2,2 — 91 stones
Cell 2,3 — 45 stones
Cell 2,4 — 24 stones
Cell 3,1 — 10 stones
Cell 3,2 — 51 stones
Cell 3,3 — 54 stones
Cell 3,4 — 22 stones
Cell 4,1 — 4 stones
Cell 4,2 — 27 stones
Cell 4,3 — 22 stones
Cell 4,4 — 15 stones

2. Activities to be undertaken in the Week ahead.

The second stage of the project will be aimed at enhancing and testing the classical machine learning pipeline and comparing it with deep learning solutions.

2.1 Model Optimization and Hyperparameter Tuning

Model parameters will also be optimized to make further improvements:

- Parameters of SVM (kernel, C value) adjustment.
- Optimization of HOG feature settings.
- Adjusting the size of the sliding window and stride.

The changing of these is to enhance the accuracy of detection and minimize errors.

2.2 Performance Evaluation

The model will be measured with the help of the standard measurements:

- Accuracy
- Precision
- Recall
- F1-score

There will be comparisons between the predicted detections and ground truth annotations.

2.3 Comparison with YOLO-Based Approach

A comparative analysis will be made on:

- HOG + SVM Classical ML approach.
- Deep learning approach (YOLO)

This comparison will bring to the fore:

- Accuracy differences

- Speed and efficiency
- Advantages and weaknesses of both methods.

2.4 Improvement in Detection Pipeline

Further improvements will involve:

- More effective preprocessors.
- Reducing false positives
- Enhancing the detection in dense stone areas.

2.5 Final Integration and System Refinement

The end system will incorporate:

- Detection
- Counting
- Visualization

The idea is to develop a strong and precise automated stone detection system to be applicable in the real world.